Amr Yasser
20230104 3

Assignment - 2

1. (a) MLE ~~when~~ for λ when α is Known:

- PDF:

$$f(x) = \begin{cases} \frac{\lambda^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$f(x;\alpha,\lambda)$

- Likelihood:

$$L(\lambda) = \prod_{i=1}^{n} \frac{\lambda^{\alpha}}{\Gamma(\alpha)} x_i^{\alpha-1} e^{-\lambda x_i} = \frac{\lambda^{n\alpha}}{[\Gamma(\alpha)]^n} \left(\prod_{i=1}^{n} x_i^{\alpha-1}\right) e^{-\lambda \sum_{i=1}^{n} x_i}$$

- Log L:

$$\ell(\lambda) = \ln L(\lambda) = n\alpha \ln \lambda - n \ln \Gamma(\alpha) + (\alpha-1)\sum_{i=1}^{n} \ln x_i - \lambda \sum_{i=1}^{n} x_i$$

- $\frac{\partial \ell}{\partial \lambda}$ w.r.t. λ = zero:

$$\frac{\partial \ell}{\partial \lambda} = \frac{n\alpha}{\lambda} - \sum_{i=1}^{n} x_i \Rightarrow \frac{n\alpha}{\hat{\lambda}} - \sum_{i=1}^{n} x_i = 0$$

$$\boxed{\hat{\lambda} = \frac{n\alpha}{\sum_{i=1}^{n} x_i}}$$

(b) $\alpha = 2$, $\boxed{\hat{\lambda} = \frac{15 \times 2}{59.25} = 0.5063291}$

(c) α and λ both unknown:

PDF, L, and $\ell(L(\alpha,\lambda))$ won't change.

also $\hat{\lambda}$ won't change either, $\hat{\lambda} = \frac{n\alpha}{\sum_{i=1}^{n} X_i}$

$$\frac{\partial \ell}{\partial \alpha} = n \ln \lambda - n\psi(\alpha) + \sum_{i=1}^{n} \ln X_i = 0,$$

$$\psi(\alpha) = \frac{d}{d\alpha} \ln \Gamma(\alpha)$$

$$\boxed{\psi(\alpha) - \ln(\alpha) = \frac{1}{n} \sum_{i=1}^{n} \ln X_i - \ln\left(\frac{1}{n} \sum_{i=1}^{n} X_i\right)}$$

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2. - PDF:

$$f(x;\alpha,\beta) = \frac{\alpha}{\beta^{\alpha}} x^{\alpha-1} e^{-(x/\beta)^{\alpha}}, \quad x \geq 0$$

- Likelihood:

$$L(\alpha,\beta) = \prod_{i=1}^{n} \frac{\alpha}{\beta^{\alpha}} X_i^{\alpha-1} e^{-(x/\beta)^{\alpha}}$$

$$= \alpha^{n} \beta^{-n\alpha} \left(\prod_{i=1}^{n} X_i^{\alpha-1}\right) e^{-\sum_{i=1}^{n} \frac{X_i^{\alpha}}{\beta^{\alpha}}}$$

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- Log-likelihood:

$$\ell(\alpha,\beta) = \ln L(\alpha,\beta) = n\ln\alpha - n\alpha\ln\beta + (\alpha-1)\sum_{i=1}^{n}\ln X_i - \beta^{-\alpha}\sum_{i=1}^{n} X_i^{\alpha}$$

- $\partial \ell$ w.r.t. each $\begin{cases}\alpha \\ \beta\end{cases}$

$$\frac{\partial \ell}{\partial \beta} = -\frac{n\alpha}{\beta} + \alpha\beta^{-\alpha-1}\sum_{i=1}^{n} X_i^{\alpha} = 0$$

$\beta^{\alpha} = \frac{1}{n}\sum_{i=1}^{n} X_i^{\alpha}$, So we can say ~~that the~~ MLE of β (as a function) of α is:

$$\hat{\beta}(\alpha) = \left(\frac{1}{n}\sum_{i=1}^{n} X_i^{\alpha}\right)^{1/\alpha}$$

$\xRightarrow{\text{So}}$ β has an explicit expression in terms of α.

$\frac{\partial \ell}{\partial \alpha} =$, first: $S_1(\alpha) = \sum_{i=1}^{n} X_i^{\alpha}$, $S_2(\alpha) = \sum_{i=1}^{n} X_i^{\alpha}\ln X_i$

$$\frac{\partial \ell}{\partial \alpha} = \frac{n}{\alpha} - n\ln\beta + \sum_{i=1}^{n}\ln X_i - \beta^{-\alpha}\sum_{i=1}^{n} X_i^{\alpha}(\ln X_i - \ln\beta)$$

From the β-equation $\Rightarrow$ $\beta^{-\alpha} S_1(\alpha) = n$

Simplify the last term:

$$\beta^{-\alpha}\sum_{i=1}^{n} X_i^{\alpha}(\ln X_i - \ln\beta) = \beta^{-\alpha} S_2(\alpha) - \ln\beta - \beta^{-\alpha} S_1(\alpha)$$

$= \beta^{-\alpha} S_2(\alpha) - n\ln\beta$. So ∂l w.r.t. α becomes:

$$\frac{\partial l}{\partial \alpha} = \frac{n}{\alpha} + \sum_{i=1}^{n} \ln X_i - \beta^{-\alpha} S_2(\alpha) = 0.$$

$\boxed{\beta^{-\alpha} = n / S_1(\alpha)}$, So: $\frac{n}{\alpha} + \sum_{i=1}^{n} \ln X_i - \frac{n S_2(\alpha)}{S_1(\alpha)}$

$\div n \longrightarrow$ $\boxed{\frac{1}{\alpha} + \frac{1}{n}\sum_{i=1}^{n} \ln X_i = \frac{\sum_{i=1}^{n} X_i^{\alpha} \ln X_i}{\sum_{i=1}^{n} X_i^{\alpha}}}$

When $\hat{\alpha}$ is found compute: $\hat{\beta} = \left(\frac{1}{n}\sum_{i=1}^{n} X_i^{\hat{\alpha}}\right)^{1/\hat{\alpha}}$

Solve $\hat{\alpha}$ numerically then compute $\hat{\beta}$.

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3. Solved in R notebook, submitted with the assignment.

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